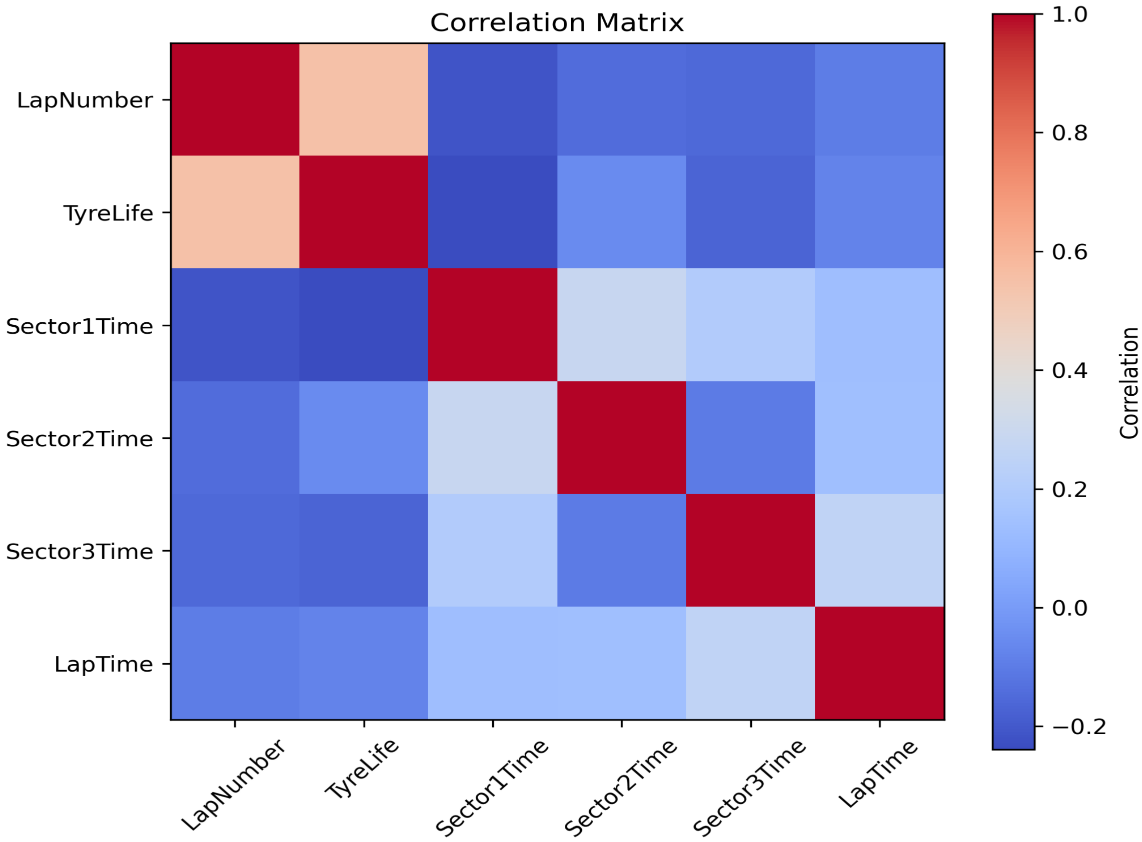
Formula 1 Lap Time Data Science Analysis

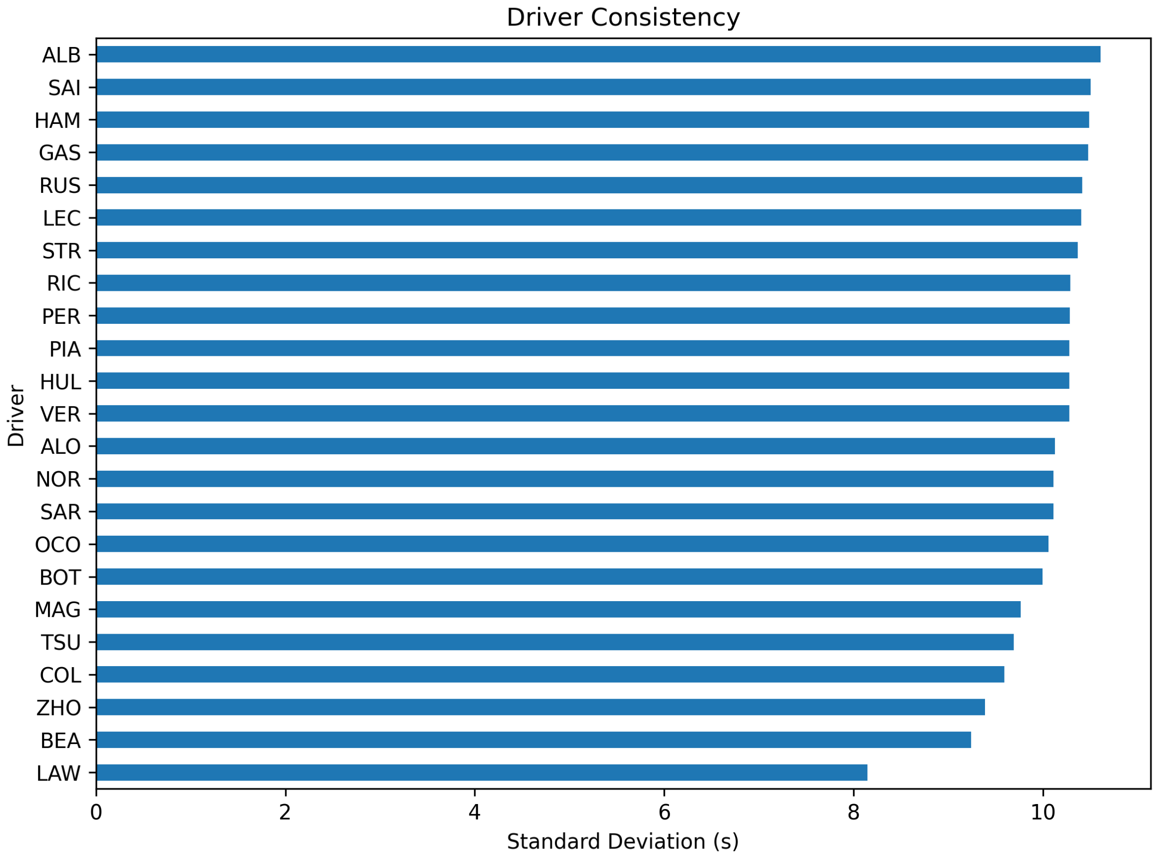
This report presents the findings from an exploratory analysis of Formula 1 lap timing data collected across the 2025 season. Each figure is accompanied by a discussion interpreting the trends shown.

Figure 1. Correlation Matrix



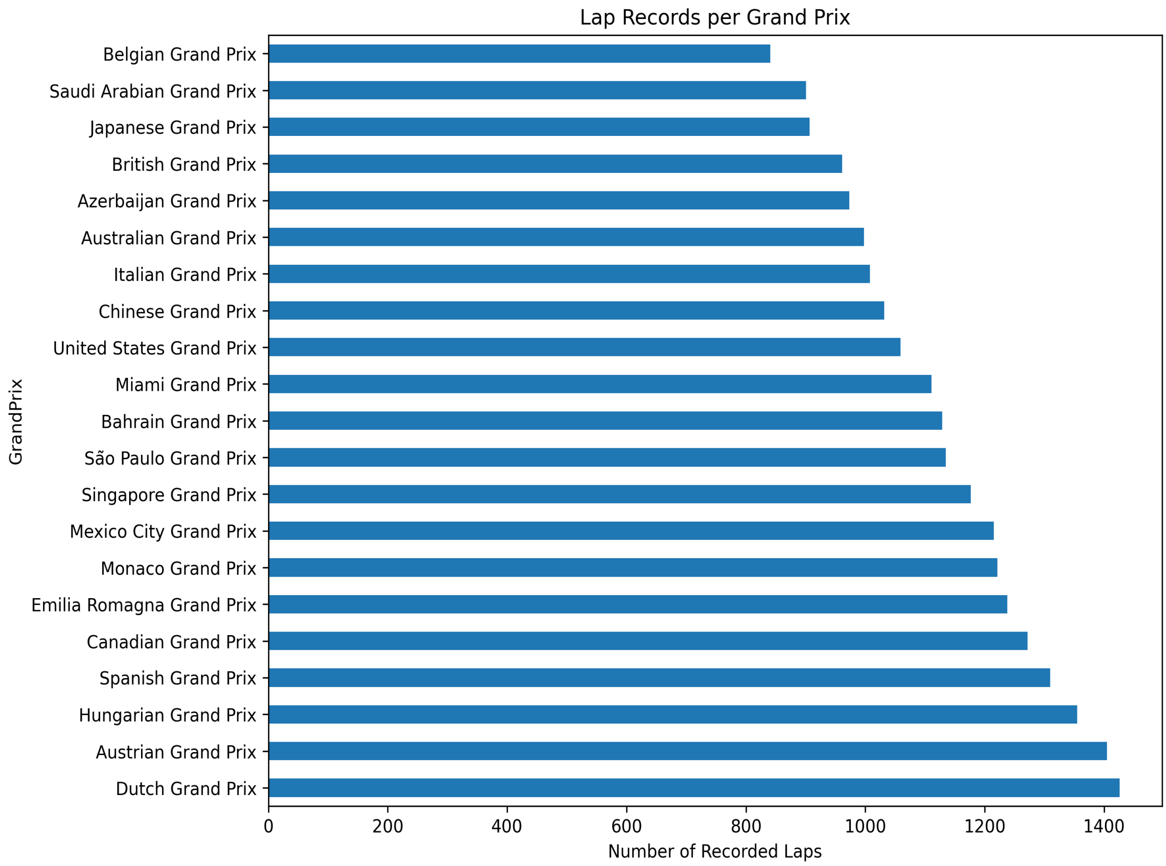
The heatmap illustrates the relationships between the numerical variables. The strongest positive correlation is between **Lap Number** and **Tyre Life**, reflecting the natural increase in tyre age as a race progresses. Sector times each contribute positively to overall lap time, although the correlations remain relatively weak, indicating that no single factor completely explains lap performance.

Figure 2. Driver Consistency



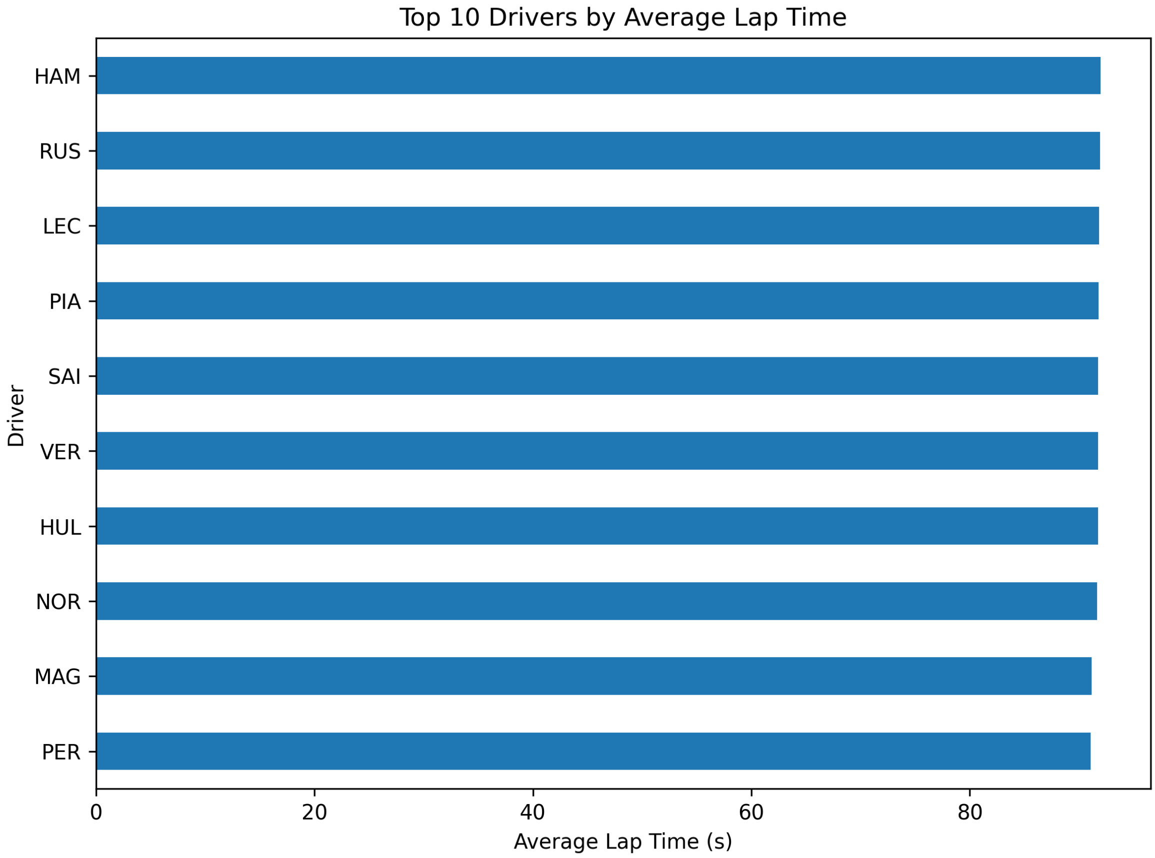
Driver consistency was measured using the standard deviation of lap times. Smaller values indicate more consistent pace throughout the season. Most drivers fall within a narrow range, suggesting similar levels of consistency across the field, although Liam Lawson records the lowest variation while Alexander Albon shows the highest.

Figure 3. Lap Records per Grand Prix



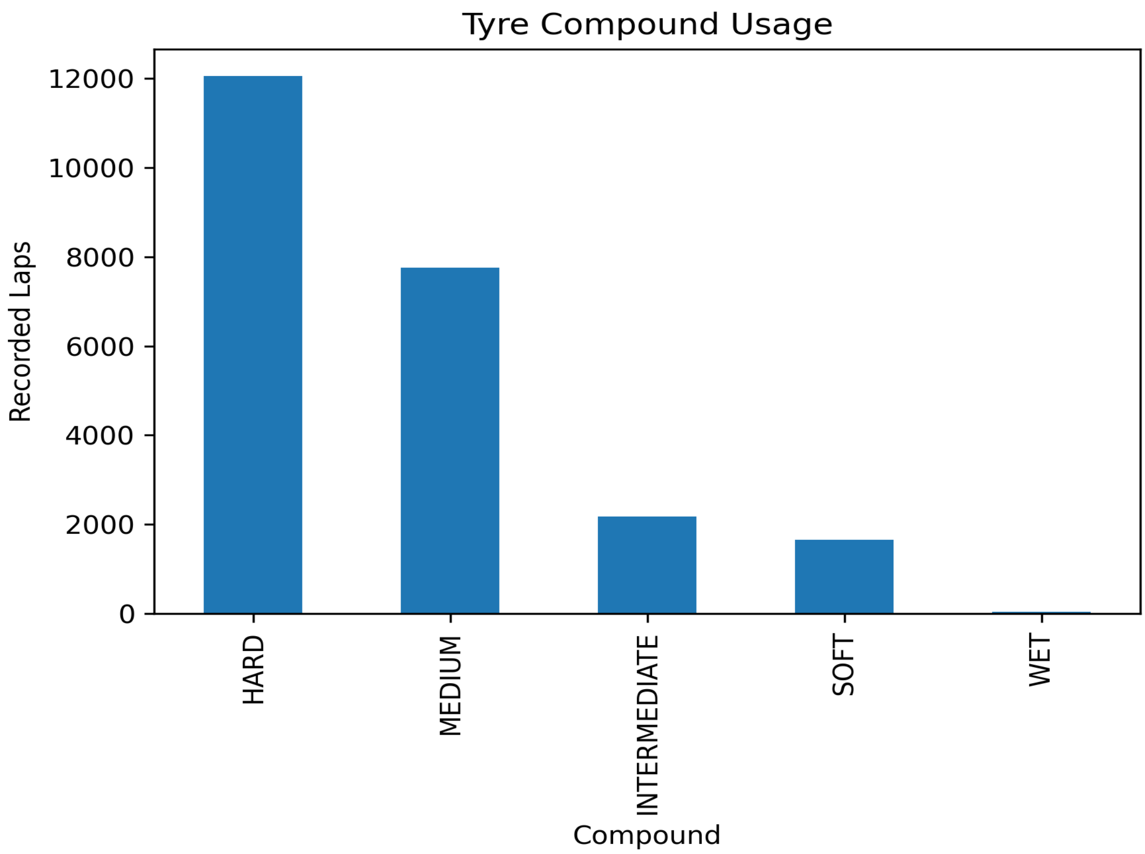
The Dutch Grand Prix contains the largest number of recorded laps, while the Belgian Grand Prix has the fewest. This uneven distribution demonstrates that some circuits contribute substantially more data, meaning statistical comparisons between races should consider differences in sample size.

Figure 4. Top 10 Drivers by Average Lap Time



The fastest ten drivers are separated by less than one second on average, highlighting the extremely competitive nature of modern Formula 1. While average lap time provides a useful summary statistic, it should be interpreted alongside variables such as tyre strategy, circuit layout and weather conditions.

Figure 5. Tyre Compound Usage



Hard tyres account for the majority of recorded laps, followed by medium tyres. Soft tyres appear much less frequently, while intermediate and wet tyres are rare. This indicates that most of the analysed race sessions were completed in dry conditions and that harder compounds were preferred for longer stints.

Conclusion

The visualisations collectively demonstrate that tyre degradation, sector performance and race strategy all contribute to lap time variation. Driver consistency is relatively similar across the grid, whereas tyre compound usage is heavily skewed towards hard and medium tyres due to dry race conditions. Although the dataset provides extensive coverage of the season, some Grand Prix contribute significantly more data than others. Future work could incorporate weather, fuel load, safety-car periods and predictive machine learning techniques to further improve the analysis.